Dynamic Aperture Control with a Transmissive Liquid Crystal Display

490480278, 500468995, 530477149, 530504892

Abstract: A small liquid crystal display, stripped of its backlight and driven by an Arduino, was placed in the light path of a digital single-lens reflex camera, in front of the lens or inside a slotted extension tube behind it. The display is a programmable aperture: any black-and-white image becomes the shape of the lens opening within a frame, and switching the whole panel between clear and black makes it a coded shutter. Four applications were tested. Through a complementary pair of coded apertures, the camera told apart two objects at 90 and 130 mm behind the lens and placed the boundary between them within 1 pixel, but only for scenes whose depth does not change vertically. Deconvolving the same pair of photographs sharpened textured regions and raised a gradient-based sharpness measure elevenfold, although ringing inflates that figure. Out-of-focus fairy lights took the shape of the displayed sprites: they matched their own pattern with a correlation of 0.81 in front of the lens, against 0.72 for a plain disc and 0.78 for the other ten patterns, so the gain specific to the displayed pattern is small, and features finer than about 1.4 mm on the panel were lost. As a flutter shutter the panel made motion blur invertible in all six runs; in dim light the coded photograph beat a short exposure by up to 2.6 dB in front of the lens and 3.1 dB behind it, but a plain exposure of the same length won in front of the lens because the code shuts the panel for half of it. Light loss through the panel, its quoted 15 to 35 ms switching time and the short working distances forced by the extension tube limited every experiment. A faster monochrome panel and blur calibrations across the whole field are the next steps.

Index Terms: Programmable aperture, coded aperture, depth from defocus, post-capture refocus, custom bokeh, flutter shutter.

1 Introduction

Computational photography pairs an unconventional optical element with processing to capture what a plain camera cannot [1]. A coded aperture replaces the round opening of the lens with a designed pattern, so each out-of-focus point records a scaled copy of that pattern, and the blur carries information that processing can decode [2]. Patterns have been designed for defocus deblurring [3], depth estimation [4] and light-field capture [5].

A cut mask holds one pattern, so each application needs its own mask and a manual swap, and no mask can change during an exposure. This project replaces the mask with a 1.8-inch colour liquid crystal display (LCD), stripped of its backlight and mounted in front of the lens or inside a slotted extension tube between the lens and the body of a Canon EOS 60D. A laptop sends each pattern, as an image or a GIF frame, to an Arduino Uno that draws it on the panel, so the aperture changes between frames, or within one exposure, with no mechanical change. Where Lab 3 photographed through fixed coded apertures, one panel here serves every pattern and doubles as a shutter. Programmable apertures have been built before from LCD and LCoS panels [6], [7], [8], [9] and reported low contrast and heavy light loss; this project works around both with long exposures and tests the panel on four applications: depth estimation, post-capture refocus, custom bokeh and flutter shutter.

1.1 Depth estimation and post-capture refocus

A lens focused at one distance images points at that distance sharply, and points nearer or farther spread into a blur that grows with their distance from the plane of focus [10]. With a round aperture that blur erases detail at some spatial frequencies. With a coded aperture the blur is a scaled copy of a known pattern, and a complementary pair of patterns loses different frequencies, so two photographs of the same scene can be compared against the measured blur at each candidate distance to recover a depth map and a sharp image [4]. Here refocus means deblurring each region with the blur measured at its estimated depth.

1.2 Custom bokeh

Bokeh is the appearance of out-of-focus light. A defocused point of light images as a scaled copy of the aperture, which is why a five-blade iris gives pentagons [11], so a pattern drawn across the iris should give lights shaped like the pattern (Section 2.3). The experiment measures how closely they follow it.

1.3 Flutter shutter

A flutter shutter makes motion blur invertible by opening and closing in a binary code during one exposure [12]. A plain long exposure smears each moving point into a uniform streak, and any stripe pattern that repeats a whole number of times along that streak averages to flat grey, so deconvolution cannot recover it. The coded streak keeps every such pattern, and deconvolution with the known code recovers a sharp image; here the LCD, switched between clear and black, is the shutter. Coded exposure helps most in dim light, where read noise dominates [13].

2 Literature Review

2.1 Depth estimation

Depth from defocus (DFD) reads distance from the size of the defocus blur [10]. One photograph cannot separate blur

- *MECH5720 Computational Imaging, The University of Sydney, 2026. Code, captures and build guide: https://github.com/j-passmore/mech5720-2026-Dynamic-Aperture-Using-LCD*

from texture, so Zhou et al. [4] optimised a pair of coded apertures with a genetic algorithm and gradient descent so that their spectra lose different frequencies. Appendix Figure 9 shows the pair this project displayed.

A thin lens of focal length F and aperture diameter D, focused at distance $\tilde{u}$, holds the sensor at $\tilde{v}$, where $1/F = 1/\tilde{u} + 1/\tilde{v}$. A point at another distance u forms a blur circle on the sensor (Appendix Figure 10) of radius

$$r = \frac{D}{2}\frac{|uF - \tilde{v}u + F\tilde{v}|}{Fu}. \quad (1)$$

An aperture with two pinholes at the ends of a diameter turns each out-of-focus point into two points on the sensor, separated by $2r$ (Appendix Figure 11); in general a defocused coded aperture images as a scaled copy of its pattern. The capture f of a scene patch at distance u is

$$f = f_0 \otimes k_u + \eta, \quad (2)$$

where f_0 is the sharp image, $\otimes$ convolution, k_u the point spread function (PSF) at that distance and η noise. Two complementary apertures give two captures of the same f_0 with different PSFs. For each candidate distance, joint deconvolution of the pair estimates f_0, the estimate is re-blurred with the same PSF pair, and the distance with the smallest residual in each local window is its depth.

2.2 Post-capture refocus

Post-capture refocus recovers a sharp image where the capture was out of focus. Levin et al. [3] designed one coded aperture whose blur can be both identified and inverted, and light-field cameras refocus after capture at the cost of resolution [14], [15]. This project follows Zhou et al. [4]: the aperture pair is deconvolved jointly with the PSF pair of each region's estimated depth, which brings regions at different distances into focus in one image.

2.3 Custom bokeh

An out-of-focus point of light images as a scaled copy of the aperture [11], so a pattern across the iris reshapes every out-of-focus light, and the width of the copy follows Equation 1 with the pattern width in place of D. On the sensor the copy of a point beyond the plane of focus is turned by 180° and that of a nearer point is upright, because the rays cross before the sensor in the first case only (Appendix Figure 17); the camera's own 180° flip of the sensor image undoes this, so in the photograph a light beyond the plane of focus shows the pattern as the photographer sees it on the panel when looking towards the scene, and a nearer light shows it turned by 180°. Programmable apertures built from LCD or LCoS panels change the pattern between frames [7], [9], and Lanman et al. reshape out-of-focus points after capture from photographs through several apertures [16]. Here the LCD sets the shape in the optics within one exposure.

2.4 Flutter shutter

Deblurring a moving object in an ordinary long exposure leaves stripes, because the box blur has zeros in its spectrum. Raskar et al. [12] opened and closed a liquid crystal shutter in front of the lens in a 52-chop code with no spectral zeros, so deconvolution with the known code recovers a sharp image. Cossairt et al. [13] showed that the flutter shutter beats a short exposure in low light, where read noise dominates, and Tendero et al. [17] proved that its gain over the best single exposure is bounded even when the motion is known. The experiment here asks whether an LCD with a quoted switching time of 15 to 35 ms, far slower than Raskar's shutter, keeps that benefit, in front of the lens and behind it at three light levels.

3 Method

A laptop sends each aperture pattern to an Arduino Uno, which draws it on the stripped LCD, and fires the camera over USB: digiCamControl on Windows for the depth and refocus runs, gphoto2 on Ubuntu for the bokeh and flutter runs. Every capture is a RAW file checked for saturation, and each experiment takes its own dark frames for black level, fixed-pattern noise and ambient light. Table 1 lists each experiment's metric and baseline.

3.1 Hardware and capture

- The aperture plane is a 1.8-inch colour LCD of 128×160 pixels at a 0.219 mm pitch [18], with its backlight, reflector and diffuser removed (Appendix Figure 8); its polariser and colour filter stay, so even its clear state absorbs most of the light.
- The camera is a Canon EOS 60D with the EF-S 18–55 mm f/3.5–5.6 zoom, at 34 mm for the depth and refocus runs and 51 to 55 mm for the bokeh and flutter runs, with the iris fully open unless an f-number is stated.
- The panel is taped flat against the front of the lens, or slid through a slot cut in the side of a 25 mm extension tube between the lens and the body, which forces close focus. The patterns were scaled to fit within the iris footprint and centred on the optical axis by eye.

3.2 Depth estimation and post-capture refocus

The complementary pair of Appendix Figure 9 estimates depth and deblurs at once, so every scene and PSF capture was taken twice, once per mask.

- The focus was set so that at least one object was sharp, though in front of the lens none was fully sharp, and each object's distance from the panel was marked.
- The scene, which held high-texture objects, was photographed once per mask at ISO 200, 2 s and f/5, and the PSF of each mask at each marked distance with the same settings, using a torch as the point source (Appendix Figure 13).
- The green channel of each capture was dark-frame corrected, and each PSF was cropped and normalised.
- For each candidate depth, the two captures were jointly deconvolved with that depth's PSF pair under a natural-image prior, the result was re-blurred with the same pair, and the residual was averaged over a local window. Each pixel takes the depth of smallest residual and its deconvolved value at that depth.

In front of the lens, five PSF pairs were measured at the image centre, 350 to 903 mm from the panel. They gave a usable refocus but no usable depth map, because the PSF varies across the field as well as with distance. Behind the lens the scene was reduced to two objects at 90 and 130 mm, as close as the tube demands, with PSF pairs at 90 to 210 mm, and the depth step was coarsened (Appendix Figure 14): large overlapping windows are each matched against the PSF pairs, the most common depth of each column is kept, and the boundary between depths is snapped to the nearest strong vertical edge. This assumes one depth per column. Refocus in both positions is the regularised joint deconvolution of the two captures with the PSF pair of each region's estimated depth.

3.3 Custom bokeh

The experiment tests whether each out-of-focus light takes the shape of the displayed pattern, scored against an open disc and against the other patterns. Appendix Figure 18 shows the layout.

- The panel showed eleven patterns: seven sprite frames, each a white character on black redrawn from its game (Mario walk 1–3 from Super Mario Bros. [19], Link attack 1–2 from The Legend of Zelda [20] and Knight walk 1–2 from Realm of the Mad God [21]), and the Zhou A, Zhou B [4], Levin [3] and Nayar masks at 13×13 cells. Each pattern spanned the iris footprint: 9.5 mm (43 panel pixels) in front of the lens and 5.5 mm (25 pixels) in the tube.
- The camera focused on a real raspberry pie, the dessert and not the Raspberry Pi computer, 216 mm from the sensor, with a string of fairy lights an estimated 1 to 1.5 m behind it, and took one 5 s photograph per pattern and position at f/5.6 and ISO 100.
- A script develops each RAW file, cuts the same square of lights out of every photograph of a position, turns the cut to face the same way as the pattern and scales each row of the figure to its brightest light. No deconvolution is applied.
- For the score, five lights per position were located on the mean of the eleven frames. For each light and frame the normalised cross-correlation (NCC) with the displayed pattern, softened by a Gaussian of 0.15 of the pattern width, was maximised over shift and over template sizes of 0.8 to 1.6 times the light's seed diameter; the orientation (one of eight flips and quarter turns) was chosen once per position by the highest mean NCC. The same light scored against an open disc and against each of the other ten patterns gives the two baselines.

3.4 Flutter shutter

The Arduino switches the LCD between clear and black in a 32-chop binary code within one exposure [12]; the code and the blur length define the blur, so no PSF is measured. Appendix Figures 20 and 21 show the setup.

- A toy car with a 95 on its side is pulled past the camera on a string during each 5 s exposure. The panel's data sheet quotes a switching time of 15 to 35 ms [22], so each chop lasts 100 ms and the code 3.2 s. Every sensor row records the same code, because the panel stays black while the shutter curtains travel.
- Each coded photograph is compared with a box exposure of the same 5 s, which collects twice the light because the code shuts 16 of the 32 chops, and with a 1/10 s impulse exposure, at f/5.6, f/11 and f/22, which set the light level, with the LCD in front of the lens and inside the tube.

TABLE 1
Experiments, metrics and baselines. E_b is the boundary error in pixels, ΔT the gradient-energy gain, NCC the normalised cross-correlation, MSE the mean squared error and PSNR the peak signal-to-noise ratio.

Experiment	Metric	Baseline
Depth estimation	Depths recovered and boundary error E_b	Measured distances and a hand-marked boundary
Post-capture refocus	Visual sharpness and gradient-energy gain ΔT	The original capture
Custom bokeh	NCC between each light and the displayed pattern	An open disc and the other ten patterns
Flutter shutter	PSNR gain in dB from the MSE against a stationary reference	Box and impulse exposures

A script stretches the 32-chop code over the blur length, sweeps blur lengths from 200 to 1200 px because the car's speed varied, and deblurs each row with the Wiener filter

$$W = H^{*} / \left(|H|^{2} + K \right), \tag{3}$$

where H is the Fourier transform of the blur and the constant K stands in for the noise-to-signal ratio, which makes the filter a Tikhonov-regularised inverse. Richardson–Lucy deconvolution [23], [24] gives a second deblur that assumes Poisson noise; the impulse photograph gets Gaussian smoothing instead. The script aligns each result to a photograph of the car standing still, matches its brightness and scores the mean squared error (MSE). Each coded and box photograph keeps its lowest MSE over K from 0.02 to 1 and 5 to 80 iterations, and the result is reported as the gain $10 \log_{10}(\mathrm{MSE}_{\mathrm{baseline}} / \mathrm{MSE}_{\mathrm{coded}})$ dB, a difference in peak signal-to-noise ratio (PSNR) that counts residual blur and ringing as noise. Figures 6 and 7 show the 40-iteration Richardson–Lucy deblurs at the best blur length.

The boundary error and the gradient-energy gain are

$$\begin{aligned} E_b &= |\hat{x}_b - x_b|, \qquad T(I) = \frac{1}{N} \sum_{x,y} |\nabla I(x,y)|^{2}, \\ \Delta T &= 100 \left(\frac{T_{\mathrm{after}}}{T_{\mathrm{before}}} - 1 \right) \%, \end{aligned} \tag{4}$$

where $\hat{x}_b$ and x_b are the estimated and hand-marked boundary columns, I the image and N its number of pixels. The bokeh NCC carries a standard deviation over light-frames and a 95 % bootstrap interval over lights. The depth, refocus and flutter metrics come from single captures without repeats.

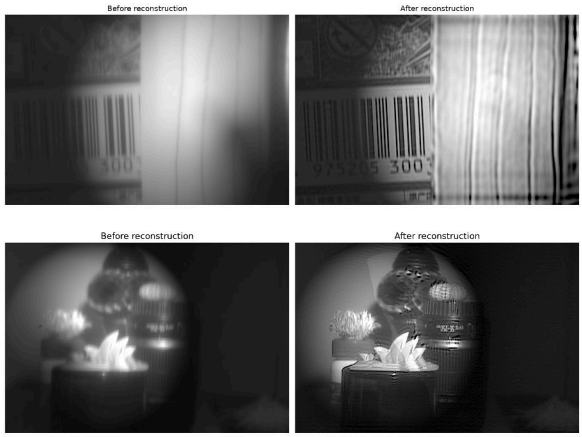


Fig. 1. Post-capture refocus: the mask 1 capture before (left) and after (right) joint deconvolution with the PSF pair of each region's estimated depth. Top: LCD behind the lens, a barcode at 130 mm and a sheet of paper at 90 mm; the barcode becomes legible and the low-texture paper rings. Bottom: LCD in front of the lens, a desk scene with a plant, an Opera House model and a lens; textured regions sharpen and ringing follows the depth boundaries.

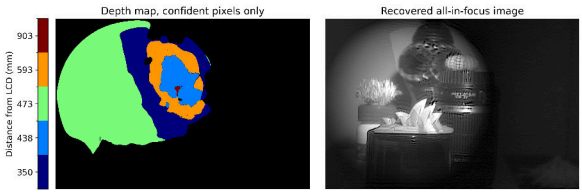


Fig. 2. Per-pixel depth estimation with the LCD in front of the lens. Left: the depth map of the desk scene of Figure 1, in mm from the panel, showing only the pixels the method trusts; the five colours are the five calibrated PSF distances. Right: the all-in-focus image from the same deblur. The estimate is dense only near the image centre, where the PSFs were measured.

4 Results

4.1 Depth estimation and post-capture refocus

Joint deconvolution recovered image detail from the coded-aperture captures in both LCD positions (Figure 1). The barcode, the Opera House model, the flower and the lens lettering become visibly sharper. The gradient energy T of the whole image rose from 0.002 to 0.023, a gain ΔT of about 1000 %, but ringing and ghost edges add gradient energy too, so this figure overstates the sharpening. The ringing marks PSF mismatch and argues for PSFs at more positions across the field. Figure 2 shows the per-pixel depth estimate in front of the lens, dense only near the image centre where the PSFs were measured. The coarser method behind the lens (Figure 3) separates the two objects and returns 90 and 130 mm, the measured distances; the objects stood at two of the five calibrated distances, so the test is whether the method picks the right candidates. The boundary snap cut the boundary error E_b from 412 to 1 px for this scene, where the strongest vertical edge is the hand-marked boundary. It is not robust: in Appendix Figure 15 a line of white text pulled the boundary the wrong way, and in Appendix Figure 16 a small object at a third depth is missed, because the method assigns one depth per column.

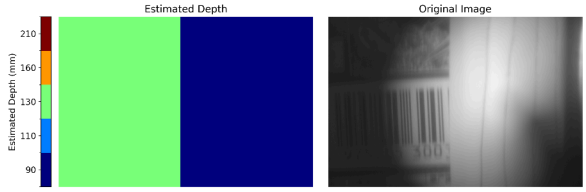


Fig. 3. Depth estimation with the LCD behind the lens by the column-wise method. Left: the estimated depth in mm from the panel. Right: the mask 1 capture, a barcode at 130 mm beside a sheet of paper at 90 mm. The method returns the two measured distances and places the boundary within 1 px of the hand-marked edge.

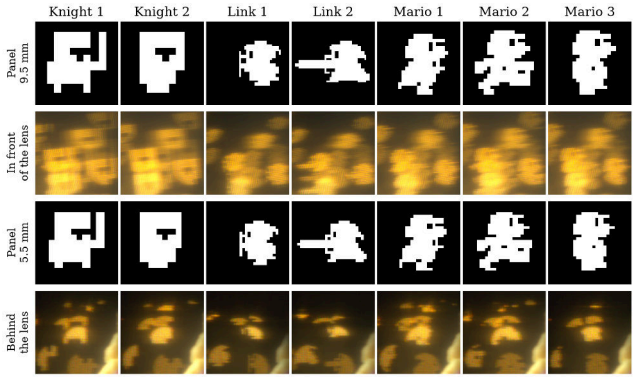


Fig. 4. Custom bokeh with seven sprite frames from Realm of the Mad God [21] (Knight), The Legend of Zelda [20] (Link) and Super Mario Bros. [19] (Mario). Rows 1 and 3: each pattern as the panel drew it (white is open) at the 9.5 mm pupil used in front of the lens and the 5.5 mm pupil used in the tube. Rows 2 and 4: the out-of-focus fairy lights photographed through that pattern, cut from the same square of every photograph of a position (1,700 and 2,200 sensor pixels wide, so the photo rows differ in scale) and scaled to the brightest light of the row. Each cut is turned 90° anticlockwise, and the front-of-lens cuts are also mirrored left to right, so that the lights face the same way as the pattern; the turn follows the panel's mounting and the mirror which face of the panel pointed at the scene (Sections 2.3 and 5.2). The mesh in every light is the panel's pixel grid.

4.2 Custom bokeh

Figure 4 shows each sprite frame above the lights it shaped, and Appendix Figure 19 the same for the four coded masks. In front of the lens every light is a readable copy of the sprite: the knight's visor slot shows in each light, but Mario's eye and button holes do not resolve. Inside the tube the lights keep sharper edges but lose most of the outline, because a far light's beam has converged below the 5.5 mm the pattern was drawn to by the time it crosses the panel, so the beam's edge, not the pattern's, bounds the light. The mesh on every light is the panel's pixel grid, with a pitch of 18.7 px in front of the lens and 42.9 px in the tube, so the pupil spreads over about 810 sensor pixels in front and 1,080 in the tube. The smaller pupil gives the larger bokeh because the panel in the tube sits in the converging beam close to the sensor, where each panel pixel covers 2.3 times as many sensor pixels. Figure 5 gives the scores. In front of the lens the NCC between a light and its displayed pattern was 0.81 ± 0.10 (mean and standard deviation over 39 light-frames from five lights), against 0.72 ± 0.09 for an open disc and 0.78 for the other ten patterns. The displayed pattern beat the disc by $+0.09$ (95 % bootstrap interval over lights $+0.06$ to $+0.11$) but the other ten patterns by only $+0.03$ ($+0.01$ to $+0.06$), and was the best match of the eleven in

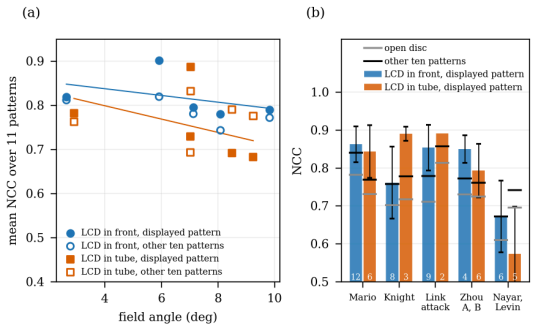


Fig. 5. Normalised cross-correlation (NCC) between each out-of-focus light and the aperture patterns. (a) Mean NCC over the eleven patterns against the light's field angle, one point per light, for the displayed pattern (filled) and the other ten patterns (open), with least-squares lines for the displayed pattern. (b) NCC by pattern group, mean and standard deviation over light-frames, with the number of light-frames at the foot of each bar; ticks mark the mean NCC against the other ten patterns (black) and against an open disc (grey).

TABLE 2
PSNR gain of the deblurred coded photograph over the box and impulse photographs in both LCD positions. Light level is the mean brightness of the reference photograph as a percentage of the sensor's full scale; a positive gain means the coded photograph scored the lower MSE. The blur sweep failed above 200 px for the box photograph marked *, so its blur and gain are uncertain. The f/22 reference in the tube is only 1.4 times darker than the f/11 reference, where two stops should give four times; whether the lamp or the car's position caused the gap was not recorded.

f-number	Light level (%)	Blur (px) coded / box	Gain vs box (dB)	Gain vs impulse (dB)
LCD in front of the lens				
f/5.6	7.8	375 / 300	−5.3	−4.4
f/11	2.1	475 / 300	−0.3	+0.3
f/22	0.6	325 / 300	−0.9	+2.6
LCD inside the extension tube				
f/5.6	5.3	500 / 400	+1.8	+3.1
f/11	1.4	375 / 200*	+2.3*	+0.1
f/22	1.0	400 / 500	−0.9	+2.5

18 % of light-frames against a chance rate of 9 %. In the tube, over 22 light-frames, the gains over the disc (+0.05, −0.03 to +0.08) and over the other ten (+0.01, −0.07 to +0.04) were indistinguishable from zero, and in neither position did the NCC fall with field angle beyond its scatter. The lights are visibly the right shape, but only marginally more like their own pattern than like any other pattern blurred to the same size, which Section 5.2 attributes to the blur.

4.3 Flutter shutter

Table 2 gives the gain of each deblurred coded photograph over the box and impulse photographs, and Figures 6 and 7 show the deblurs.

LCD in front of the lens. The gain over the impulse photograph rises from −4.4 dB at f/5.6 to +2.6 dB at f/22 as the light falls. The box photograph beats the coded photograph at every f-number, by 0.3 to 5.3 dB; at f/11 the coded photograph carries 1.6 times the blur of the box and still comes within 0.3 dB of it, and at f/22 its deblur keeps sharper strokes than the box deblur but more grain (Appendix Figure 22).

LCD inside the extension tube. The coded photograph beats the box photograph by 1.8 dB at f/5.6 and by an uncertain 2.3 dB at f/11, and loses by 0.9 dB at f/22, the same margin as in front of the lens. It beats the impulse photograph by 3.1 dB at f/5.6 and 2.5 dB at f/22 and ties within 0.1 dB at f/11. The f/5.6 and f/11 box photographs sat 1,060 to 2,050 px from their reference spots, which favours the coded photograph (Section 5.3); the f/22 box sat closer than the coded photograph but carried 1.25 times its blur. Reference position also biases the impulse comparison against the coded photograph at f/11 and towards it at f/22.

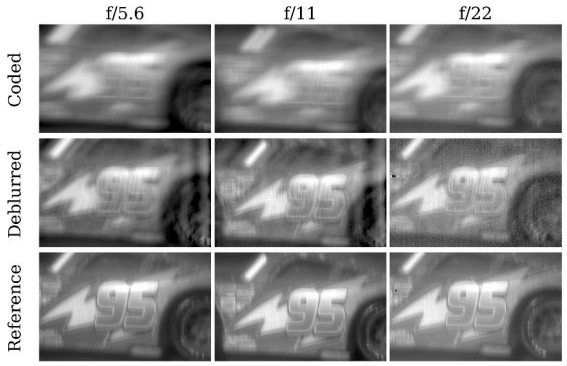


Fig. 6. Coded, deblurred and reference images of the 95 on the toy car at f/5.6, f/11 and f/22 with the LCD in front of the lens. Each deblur uses 40 Richardson–Lucy iterations at the blur length in Table 2.

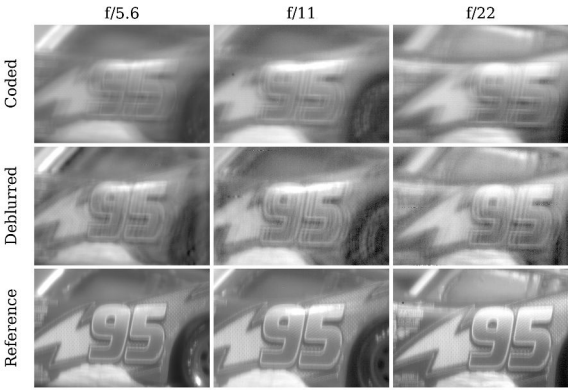


Fig. 7. Coded, deblurred and reference images of the 95 at f/5.6, f/11 and f/22 with the LCD inside the extension tube. Each deblur uses 40 Richardson–Lucy iterations at the blur length in Table 2.

Figures 6 and 7 show the deblur restoring the outline of the 95 in all six runs, with faint repeated edges around each stroke. Grain grows as the light falls and dominates the f/22 deblur in front of the lens. Inside the tube the best deblur lowers the coded photograph's own MSE by 3.5 dB at f/5.6, 1.2 dB at f/11 and 1.8 dB at f/22, a gain that car position cannot inflate, since the coded photograph and its deblur share one position.

5 Limitations and Pitfalls

The panel and the extension tube limit every experiment. The LCD's polariser and colour filter absorb most of the light, which forced exposures of 2 to 5 s, and its glass adds aberration and vignetting: photographs through the panel in the tube were crisp, but none through the panel in front of the lens. The tube forces close focus, so every scene behind the lens spans a few centimetres. The panel's transmission when clear and its leakage when black were not measured.

5.1 Depth estimation and post-capture refocus

- **Spatially varying PSFs:** the panel sits a few centimetres from the pupil, so an off-axis beam crosses it displaced by a distance comparable to the pupil width and sees a different part of the pattern; the PSF therefore varies across the field as well as with distance, but was measured only at the image centre, so the per-pixel method fails away from the centre (Figure 2).
- **One depth per column:** the column-wise method assumes that depth does not change vertically, and it misses objects that break that assumption (Appendix Figure 16).
- **Ringing:** a region deconvolved with the wrong PSF rings and grows ghost edges (Figure 1), which inflates the gradient-energy gain.

5.2 Custom bokeh

- **Blur removes detail:** the fit needed a Gaussian softening with a standard deviation of 0.15 of the pattern width, 1.4 mm on the panel in front of the lens and 0.8 mm in the tube, so the sprites' eye and button holes are lost. Diffraction by the pixel grid, which Dietz [9] reports for LCD apertures, the size of each light and the lens all contribute, and the data do not separate them.
- **Weak controls:** an open disc is a poor baseline, because the other ten patterns also beat it (by $+0.06$ in front of the lens), and no open-disc or card frame was photographed, so the NCC of an ideal aperture through these optics is unknown.
- **Few lights, no repeats:** five lights per position passed the contrast rule, in 39 of 55 light-frames in front of the lens and 22 of 55 in the tube, from one capture per pattern; intervals are over lights.
- **Orientation by eye:** the crops were turned and mirrored to match the pattern, consistent with the display face pointing at the scene in front of the lens and at the sensor in the tube, but the eight orientations differ by only 0.02 in mean NCC in front, so the data do not fix it. The distances to the lights were not recorded, so Equation 1 cannot predict the copy size.

5.3 Flutter shutter

- **Slow LCD:** the panel's quoted switching time of 15 to 35 ms was not measured; it forced 100 ms chops and a 5 s exposure. The box comparison is at equal time, not equal light.
- **Uneven car speed:** the car's speed, 100 to 160 px/s from the coded blurs, could not be held steady for the whole 3.2 s code, so only the f/22 pair in front of the lens has coded and box blurs within 10 % of each other; the other five pairs differ by 25 to 88 %.
- **Reference position:** the MSE penalises a photograph taken away from the reference spot, because the reflections and lamp light on the car change along the track, and the blur length, K and iteration count were chosen by the lowest MSE against that reference, which a camera in the field does not have.

6 Conclusions and Future Work

6.1 Depth estimation and post-capture refocus

The LCD works as a coded aperture pair for depth from defocus, as a proof of concept. The per-pixel method gave an unreliable depth map because the PSFs were calibrated only at the image centre; one depth per column recovered the two measured distances behind the lens. Joint deconvolution recovered detail in both positions, because the pair carries complementary information in the Fourier domain (Appendix Figure 12), but regions deconvolved with the wrong PSF rang and amplified noise. With two more months the group would measure PSFs across the field and at more distances, or move the panel to the pupil plane so that one PSF serves the field, improve the regularisation, and mount the panel behind the lens without the tube so that scenes can be farther away.

6.2 Custom bokeh

The LCD shaped every out-of-focus light into the displayed sprite, most clearly in front of the lens, where the knight's visor slot showed in every light. The quantitative test is sobering: the lights matched their own pattern only marginally better than the other ten, by $+0.03$ NCC in front of the lens and $+0.01$ in the tube, because a blur of about 1.4 mm on the panel removes the detail that tells the sprites apart; features wider than about 4 mm should survive. Next the group would photograph an open disc and a laser-cut card of each shape as baselines, record the distance to each light, use more lights with repeats, and try a monochrome panel, which removes the colour filter but keeps the pixel grid.

6.3 Flutter shutter

The LCD works as a flutter shutter in both positions, and deconvolution recovered the 95 in all six runs. In front of the lens the gain over the 1/10 s impulse photograph rose from -4.4 dB at f/5.6 to $+2.6$ dB at f/22 as the light fell, the low-light advantage that Cossairt et al. [13] predict, while the box photograph won at all three f-numbers, since the 16 shut chops cost the coded photograph half its light. Inside the tube the coded photograph beat the box by 1.8 dB at f/5.6 and 2.3 dB at f/11, but reference position favoured it in both runs, so this margin needs a repeat with matched reference spots. The slow LCD limited the system most: its quoted 15 to 35 ms switching time forced 100 ms chops and a 3.2 s code. A panel switching within 3.5 ms would allow 10 ms chops and a 0.32 s code for targets ten times faster.

Acknowledgements and use of AI tools

Generative AI was used in this assignment, as University policy requires us to disclose. Claude Opus 5.5 and Claude Fable 5.1 (Anthropic) and ChatGPT-6 (OpenAI) suggested and debugged code, drafted figure scripts, helped format LaTeX and proofread the text; Claude Fable 5.1 also assembled the GitHub repository to the handout's specification. We built the hardware, took every photograph, ran the experiments, checked every number against the captured data and reviewed every AI output before using it, and we take responsibility for all content in this report and the repository.

REFERENCES

[1] A. Chiranjan, "Programmable aperture photography: An investigation into applications and methods," Master's thesis, University of Cape Town, 2018. [Online]. Available: https://hdl.handle.net/11427/29456

[2] T. M. Cannon and E. E. Fenimore, "Coded aperture imaging: Many holes make light work," *Optical Engineering*, vol. 19, no. 3, pp. 283–289, 1980. [Online]. Available: https://doi.org/10.1117/12.7972511

[3] A. Levin, R. Fergus, F. Durand, and W. T. Freeman, "Image and depth from a conventional camera with a coded aperture," *ACM Trans. Graph.*, vol. 26, no. 3, Jul. 2007, Art. no. 70, doi: 10.1145/1276377.1276464.

[4] C. Zhou, S. Lin, and S. K. Nayar, "Coded aperture pairs for depth from defocus and defocus deblurring," *International Journal of Computer Vision*, vol. 93, no. 1, pp. 53–72, 2011. [Online]. Available: https://www.cs.columbia.edu/CAVE/publications/pdfs/Zhou_IJCV11.pdf

[5] C.-K. Liang, T.-H. Lin, B.-Y. Wong, C. Liu, and H. H. Chen, "Programmable aperture photography: Multiplexed light field acquisition," *ACM Trans. Graph.*, vol. 27, no. 3, Aug. 2008, Art. no. 55, doi: 10.1145/1360612.1360654.

[6] A. Zomet and S. K. Nayar, "Lensless imaging with a controllable aperture," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, New York, NY, USA, Jun. 2006, pp. 339–346, doi: 10.1109/CVPR.2006.175.

[7] H. Nagahara, C. Zhou, T. Watanabe, H. Ishiguro, and S. K. Nayar, "Programmable aperture camera using LCoS," in *Proc. 11th Eur. Conf. Comput. Vis. (ECCV)*, ser. Lecture Notes in Computer Science, vol. 6316, Heraklion, Greece, Sep. 2010, pp. 337–350, doi: 10.1007/978-3-642-15567-3_25.

[8] O. Bimber, H. Qureshi, A. Grundhöfer, M. Grosse, and D. Danch, "Adaptive coded aperture photography," in *Proc. 7th Int. Symp. Vis. Comput. (ISVC)*, ser. Lecture Notes in Computer Science, vol. 6938, Las Vegas, NV, USA, Sep. 2011, pp. 54–65, doi: 10.1007/978-3-642-24028-7_6.

[9] H. Dietz, "Programmable liquid crystal apertures and filters for photographic lenses," *IS&T Electron. Imag.*, vol. 33, no. 7, pp. 120-1–120-7, 2021, doi: 10.2352/ISSN.2470-1173.2021.7.ISS-120.

[10] A. P. Pentland, "A new sense for depth of field," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-9, no. 4, pp. 523–531, Jul. 1987, doi: 10.1109/TPAMI.1987.4767940.

[11] H. H. Nasse, "Depth of field and bokeh," Carl Zeiss AG, Camera Lens Division, Tech. Rep., Mar. 2010. [Online]. Available: https://lenspire.zeiss.com/photo/app/uploads/2018/04/Article-Bokeh-2010-EN.pdf

[12] R. Raskar, A. Agrawal, and J. Tumblin, "Coded exposure photography: Motion deblurring using fluttered shutter," *ACM Trans. Graph.*, vol. 25, no. 3, pp. 795–804, Jul. 2006, doi: 10.1145/1141911.1141957.

[13] O. Cossairt, M. Gupta, and S. K. Nayar, "When does computational imaging improve performance?" *IEEE Trans. Image Process.*, vol. 22, no. 2, pp. 447–458, Feb. 2013, doi: 10.1109/TIP.2012.2216538.

[14] R. Ng, M. Levoy, M. Brédif, G. Duval, M. Horowitz, and P. Hanrahan, "Light field photography with a hand-held plenoptic camera," Stanford Univ., Stanford, CA, USA, Tech. Rep. CSTR 2005-02, Apr. 2005. [Online]. Available: https://graphics.stanford.edu/papers/lfcamera/

[15] A. Veeraraghavan, R. Raskar, A. Agrawal, A. Mohan, and J. Tumblin, "Dappled photography: Mask enhanced cameras for heterodyned light fields and coded aperture refocusing," *ACM Trans. Graph.*, vol. 26, no. 3, Jul. 2007, Art. no. 69, doi: 10.1145/1276377.1276463.

[16] D. Lanman, R. Raskar, and G. Taubin, "Modeling and synthesis of aperture effects in cameras," in *Proc. 4th Int. Symp. Comput. Aesthet. Graph. Vis. Imag. (CAe)*, Lisbon, Portugal, Jun. 2008, pp. 81–88, doi: 10.2312/COMPAESTH/COMPAESTH08/081-088.

[17] Y. Tendero, J.-M. Morel, and B. Rougé, "The flutter shutter paradox," *SIAM J. Imaging Sci.*, vol. 6, no. 2, pp. 813–847, 2013, doi: 10.1137/120880665.

[18] Adafruit Industries, "1.8-inch color TFT LCD display with MicroSD card breakout, ST7735R, product 358," Product page. Accessed: Oct. 1, 2026. [Online]. Available: https://www.adafruit.com/product/358

[19] Nintendo, "Super Mario Bros." Video game, Nintendo Family Computer, Sep. 1985.

[20] Nintendo, "The Legend of Zelda," Video game, Family Computer Disk System, Feb. 1986.

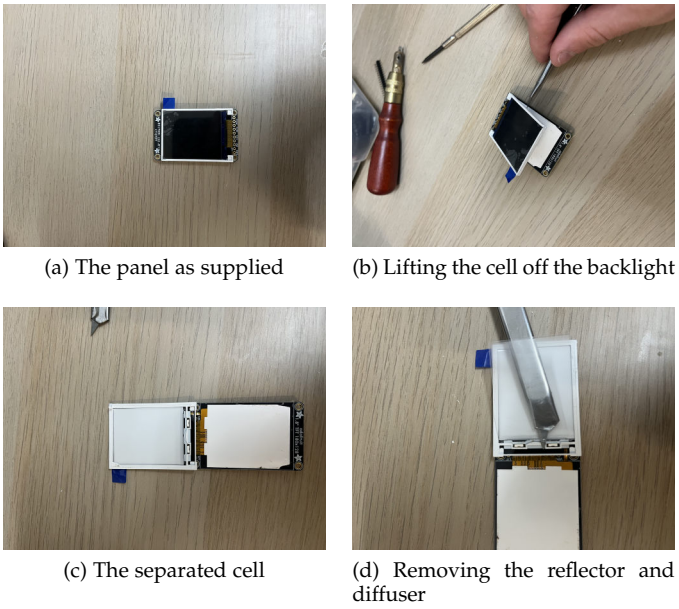
[21] Wild Shadow Studios and Spry Fox, "Realm of the Mad God," Video game, browser, public beta, now published by DECA Games, 2010.

[22] Jiandakangjing, "JD-T1800 1.8-inch TFT LCD module specification," Data sheet. Accessed: Oct. 1, 2026. [Online]. Available: https://cdn-shop.adafruit.com/datasheets/JD-T1800.pdf

[23] W. H. Richardson, "Bayesian-based iterative method of image restoration," *J. Opt. Soc. Am.*, vol. 62, no. 1, pp. 55–59, Jan. 1972, doi: 10.1364/JOSA.62.000055.

[24] L. B. Lucy, "An iterative technique for the rectification of observed distributions," *Astron. J.*, vol. 79, no. 6, pp. 745–754, Jun. 1974, doi: 10.1086/111605.

APPENDIX A
EXPERIMENTAL SETUP



(a) The panel as supplied (b) Lifting the cell off the backlight

(c) The separated cell (d) Removing the reflector and diffuser

Fig. 8. Stripping the Adafruit 1.8-inch LCD so that light passes through it: (a) the panel as supplied, (b) the cell lifted off its backlight, (c) the bare cell, and (d) the reflector and diffuser sheets removed. The polariser and colour filter stay on the cell.

APPENDIX B
DEPTH ESTIMATION AND POST-CAPTURE REFOCUS



Fig. 9. The 33×33 coded aperture pair of Zhou, Lin and Nayar [4], white where light passes. The depth and refocus runs displayed these masks; the bokeh runs displayed 13×13 versions.

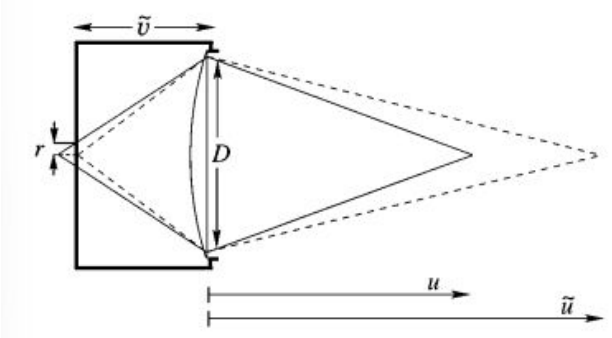


Fig. 10. A thin lens of aperture D focused at $\tilde{u}$ with the sensor at $\tilde{v}$. A point at another distance u images to a blur circle of radius r on the sensor, Equation 1.

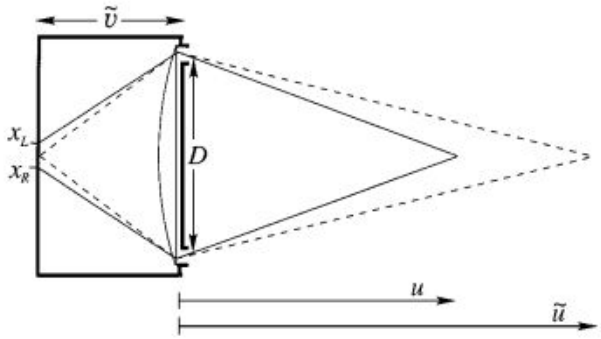


Fig. 11. The same lens with two pinholes at the ends of a diameter. An out-of-focus point lands at two positions on the sensor, separated by the blur circle diameter $2r$ of the open lens, which is how a coded aperture encodes depth.

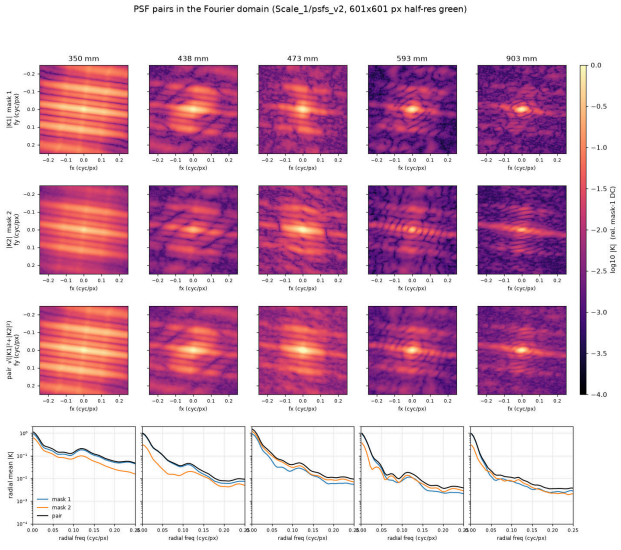


Fig. 12. Magnitude spectra of the measured PSFs with the LCD in front of the lens, for mask 1 (top row), mask 2 (middle row) and the pair combined (third row) at the five calibrated distances, with the radial means below. Each mask loses different frequencies, so the pair together covers the spectrum, which is what joint deconvolution exploits.

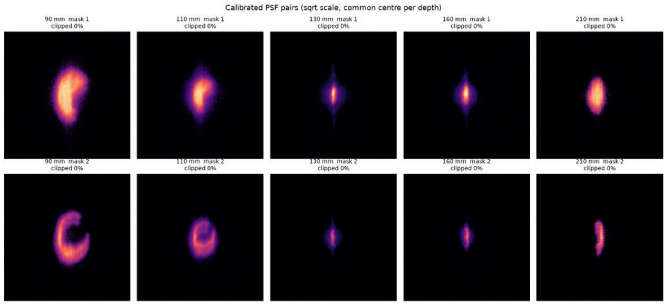


Fig. 13. The PSF pairs measured behind the lens: mask 1 (top) and mask 2 (bottom) at 90, 110, 130, 160 and 210 mm from the panel, cropped about a common centre and shown on a square-root scale. No pixel clipped.

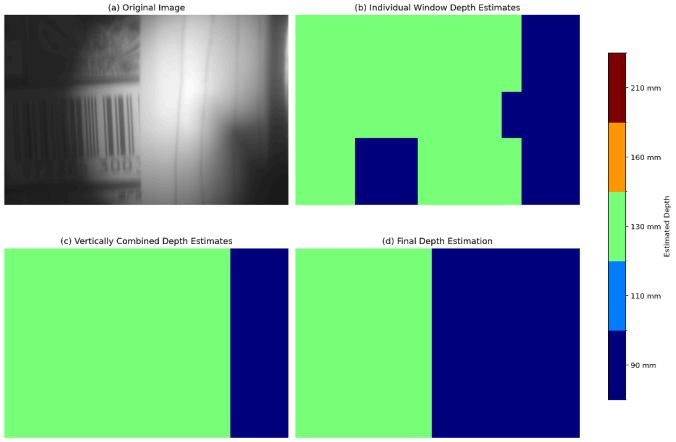


Fig. 14. The stages of the column-wise depth method behind the lens: (a) the mask 1 capture, (b) the depth chosen by each overlapping window, (c) the windows combined vertically by the most common depth of each column, and (d) the boundary snapped to the nearest strong vertical edge.

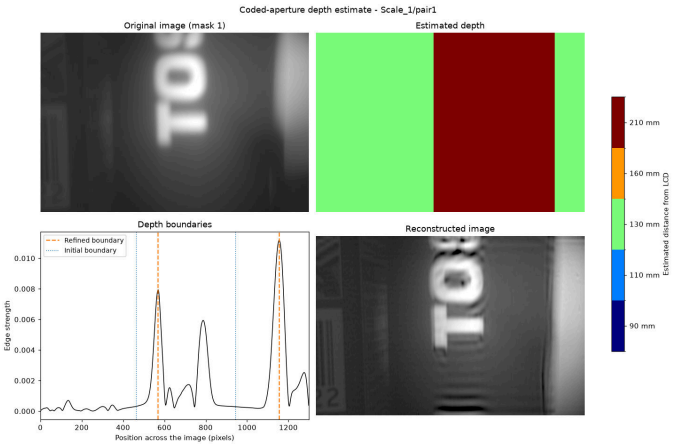


Fig. 15. A scene where the boundary snap reduced the accuracy. Top: the mask 1 capture and the estimated depth. Bottom: the edge strength across the image with the initial and refined boundaries, and the reconstruction. The bright white text drew the refined boundaries towards its edges.

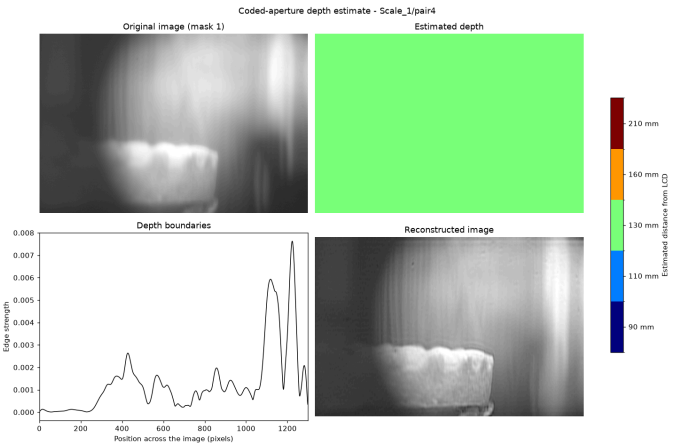


Fig. 16. A scene where the column-wise method misses a small object at a different depth and assigns the whole image one depth, because it keeps one depth per column. Panels as in Figure 15.

Appendix C
Custom bokeh

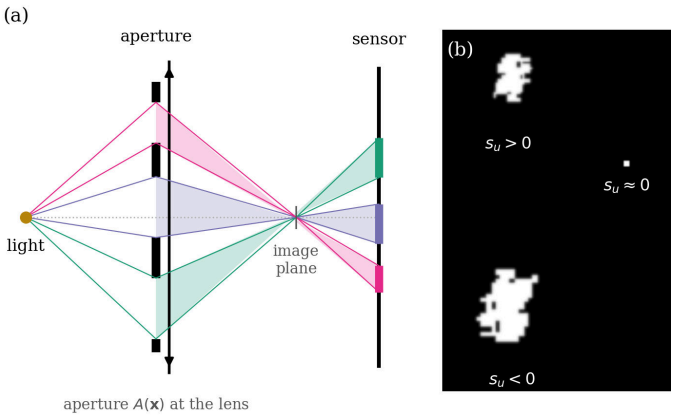


Fig. 17. How an aperture pattern becomes bokeh. (a) A light beyond the plane of focus converges in front of the sensor, so the rays through the open parts of the aperture cross and land inverted on the sensor, each coloured beam in its own strip. (b) Simulated bokeh of three lights with the Mario sprite as the aperture, at illustrative sizes: a near light ($s_u > 0$) gives an upright copy on the sensor, a light on the plane of focus a point, and a far light ($s_u < 0$) a copy turned by $180°$. The camera's $180°$ image flip reverses both orientations in the final photograph.

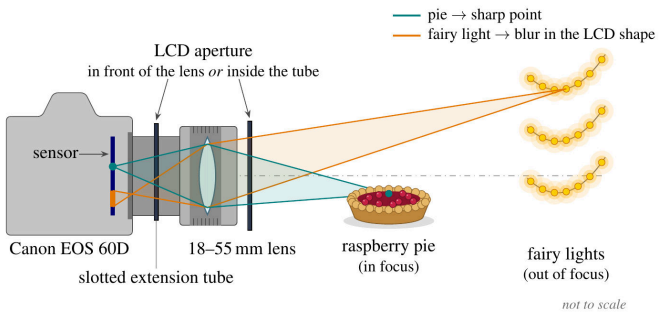


Fig. 18. The custom bokeh setup from the side. The camera focuses on a raspberry pie, a real dessert, which images to a sharp point; each fairy light behind the pie blurs into the shape of the LCD pattern. The LCD sits in front of the lens or inside the slotted extension tube. Not to scale.

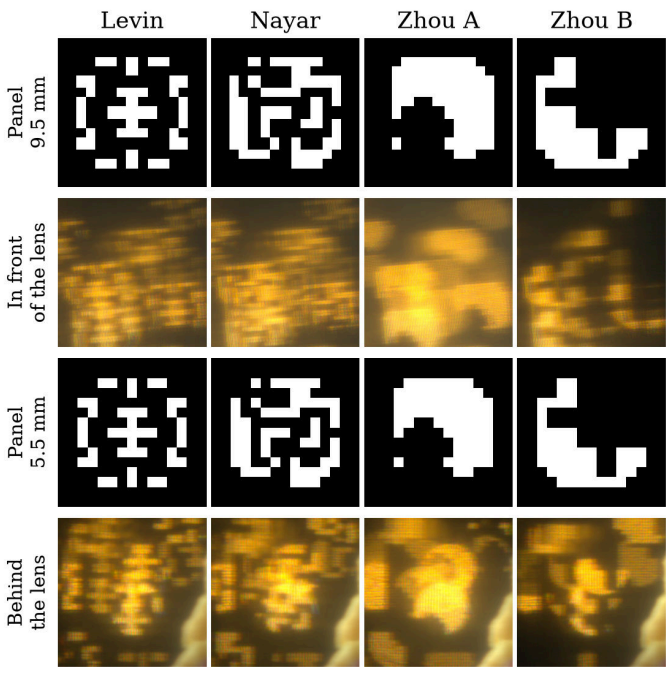


Fig. 19. Custom bokeh with the four coded masks, laid out as in Figure 4: the Levin, Nayar, Zhou A and Zhou B masks as the panel drew them at the 9.5 mm pupil (row 1) above the lights they shaped in front of the lens (row 2), and at the 5.5 mm pupil (row 3) above the lights in the tube (row 4). The cuts are turned and mirrored as in Figure 4.

Appendix D
Flutter shutter

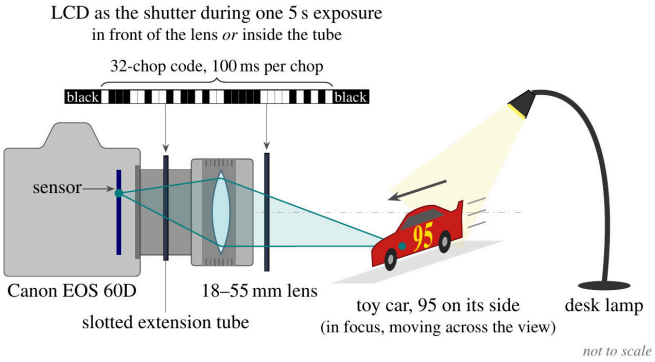


Fig. 20. The flutter shutter layout from the side. The LCD sits in front of the lens or inside the slotted extension tube and plays the 32-chop code while the toy car crosses the view during one 5 s exposure. Not to scale.



Fig. 21. The flutter shutter setup with the LCD taped to the front of the lens, the lamp on a direct current supply and the toy car on its track.

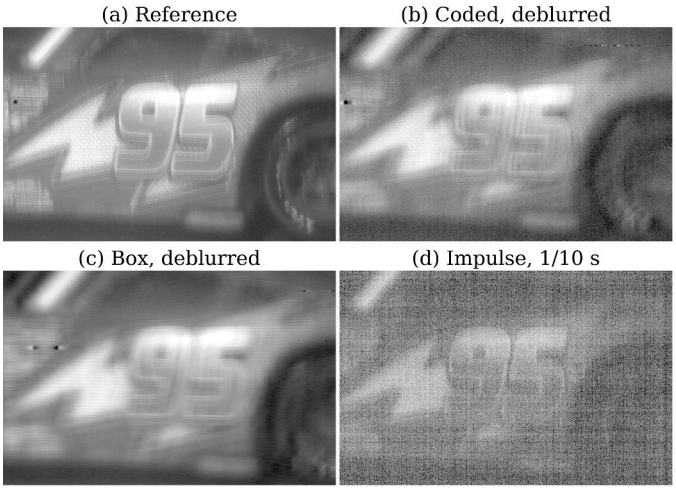


Fig. 22. The reference (a), the deblurred coded photograph (b), the deblurred box photograph (c) and the 1/10 s impulse photograph (d) at f/22 with the LCD in front of the lens. A Wiener filter with $K = 0.05$ deblurs (b) and (c). The box photograph comes from a second run.